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MLA0201-FUNDAMENTAL OF MACHINE LEARNING

Annexure A
Mock Interview

Q1. “Suppose you are working on an industrial monitoring system. How would you use Bayesian Networks to predict machine failures?” (5 Marks – 8 Mins)

Q2. “Imagine you are asked in an interview to design an image denoising solution. How would Markov Random Fields help model neighborhood pixel dependencies?” (5 Marks – 8 Mins)

Q3. “If a company wants to build a speech recognition system, how would you apply a Hidden Markov Model to process sequential audio signals?” (5 Marks – 8 Mins)

Q4. “During a technical interview, how would you explain the use of Conditional Random Fields for sequence labeling tasks such as part-of-speech tagging?” (5 Marks – 8 Mins)

Q5. “Suppose an interviewer asks how conditional independence simplifies probabilistic inference. How would you justify it with a graphical model example?” (5 Marks – 8 Mins)

Q6. “In a healthcare diagnosis problem, how would you compare directed graphical models and undirected graphical models for representing symptom relationships?” (5 Marks – 8 Mins)

Q7. “If patient symptom data is partially missing, how would you perform Bayesian inference to still make an accurate diagnosis?” (5 Marks – 8 Mins)

Q8. “How would you use a Hidden Markov Model to analyze customer purchase sequences in an e-commerce recommendation system?” (5 Marks – 8 Mins)

Q9. “In an NLP interview task, how would you explain why Conditional Random Fields often outperform Hidden Markov Models in named entity recognition?” (5 Marks – 8 Mins)

Q10. “Suppose you are asked to improve fraud detection. How would graphical model learning and generalization enhance prediction accuracy?” (5 Marks – 8 Mins)

Q1. Bayesian Network for Predicting Machine Failures

A **Bayesian Network (BN)** represents relationships between machine conditions using a graph of nodes and probabilities.

Steps:

1. Create nodes such as **Temperature, Vibration, Pressure, Lubrication** and **Machine Failure**.
2. Connect related variables using directed arrows.
3. Assign probability values based on historical machine data.
4. When sensor values are received, use **Bayesian inference** to calculate the probability of failure.
5. If the failure probability is high, generate an **early warning** for maintenance.

Example:
High temperature + high vibration → high probability of machine failure.

Advantage: It can combine uncertain sensor information and predict failures before they occur.

Q2. MRF for Image Denoising

A **Markov Random Field (MRF)** models relationships between neighboring pixels in an image.

Steps:

1. Represent each pixel as a node.
2. Connect each pixel to its neighboring pixels.
3. Define an energy function based on pixel values.
4. Noisy pixels are corrected by considering the values of their neighbors.
5. Minimize the total energy to obtain a cleaner image.

Example:
If most neighboring pixels are white but one pixel is black because of noise, the MRF can change the noisy pixel to white.

Advantage: MRF preserves local image structure while reducing noise.

Q3. HMM for Speech Recognition

A **Hidden Markov Model (HMM)** is useful because speech is a sequence of changing audio signals.

Steps:

1. Divide the speech signal into small frames.
2. Extract features such as **MFCC** from each frame.
3. Treat phonemes or speech sounds as **hidden states**.
4. Use transition probabilities to represent how one sound follows another.
5. Use emission probabilities to relate audio features to hidden states.
6. Find the most likely sequence of words using algorithms such as

Viterbi. **Example:**
Audio signal → phonemes → words → sentence.

Advantage: HMM handles sequential and time-dependent speech data effectively.

Q4. CRF for Part-of-Speech Tagging

A **Conditional Random Field (CRF)** is a probabilistic model used to assign labels to a sequence of words.

Steps:

1. Input a sentence as a sequence of words.
2. Extract features such as the word, neighboring words, suffix, and capitalization.
3. Assign labels such as **Noun, Verb, Adjective**, etc.
4. CRF considers relationships between neighboring labels.
5. Select the label sequence with the highest probability.

Example:
"Dogs run fast" → **Noun** → **Verb** → **Adverb**

Advantage: CRF considers the whole sequence and surrounding context, improving tagging accuracy.

Q5. Conditional Independence in Probabilistic Inference

Conditional independence means that two variables become independent when another variable is known.

Consider the graphical model:

Rain → Wet Grass ← Sprinkler

Here, Rain and Sprinkler affect Wet Grass.

If we know the condition of the relevant variable, some probabilities can be simplified instead of calculating every possible combination.

For example:

P(Rain, Sprinkler, Wet Grass)
can be factorized as:

P(Rain) × P(Sprinkler) × P(Wet Grass | Rain, Sprinkler)

Advantage: Conditional independence reduces the number of calculations and makes probabilistic inference faster and easier.

Q6. Directed vs Undirected Graphical Models

Directed Model (Bayesian Network)	Undirected Model (MRF)
Uses directed arrows	Uses undirected edges
Represents cause-effect relationships	Represents relationships between variables
Uses conditional probabilities	Uses compatibility/energy functions
Good for diagnosis and prediction	Good for spatial relationships
Example: Disease → Symptoms	Example: neighboring image pixels

In healthcare:
A Bayesian Network can represent **Disease → Fever → Symptoms**. An MRF can represent dependencies between related symptoms without specifying a direction.

Conclusion: Bayesian Networks are useful for causal reasoning, while MRFs are useful for modeling mutual relationships.

Q7. Bayesian Inference with Missing Patient Data

When some patient symptoms are missing, **Bayesian inference** can still estimate the probability of diseases.

Steps:

1. Build a Bayesian Network containing diseases, symptoms, age, test results, etc.

2. Enter the available patient information.
3. Leave missing symptoms as unknown rather than assuming a value.
4. Calculate the posterior probability of each disease.
5. Select the disease with the highest probability.

Example:
If fever is known but cough information is missing, the system can still calculate:

P(Disease | Fever)

using the available evidence.

Advantage: Bayesian Networks can handle incomplete and uncertain medical data.

Q8. HMM for Customer Purchase Sequences

An **HMM** can identify hidden customer interests from their purchase history.

Steps:

1. Collect customer purchase sequences.
2. Treat customer interests as **hidden states**, such as Electronics, Clothing, or Sports.
3. Treat purchases as **observations**.
4. Calculate transition probabilities between customer interests.
5. Use the current purchase history to predict the next likely interest.
6. Recommend products related to the predicted interest.

Example:
Phone → Earphones → Smartwatch
may indicate a hidden state of **Electronics interest**.

Advantage: HMM can model changes in customer interests over time.

Q9. Why CRF Often Outperforms HMM in Named Entity Recognition

CRFs often perform better than HMMs because they can use **many overlapping features** of the input sequence.

HMM	CRF
Generative model	Discriminative model
Models observations and labels jointly	Directly models labels given observations
Makes stronger independence assumptions	Fewer independence assumptions
Limited feature flexibility	Can use many features
Less effective with complex context	Handles context better

Example:
In “Apple released a new iPhone,” CRF can use capitalization, neighboring words, and word patterns to identify **Apple** as an organization.

Conclusion: CRFs are often better for NER because they consider rich contextual features and dependencies between neighboring labels.

Q10. Graphical Model Learning and Generalization for Fraud Detection

Graphical models can represent relationships between different variables involved in fraud.

Steps:

1. Create variables such as **Transaction Amount, Location, Time, Device, User Behavior** and **Fraud**.
2. Build a Bayesian Network or other graphical model.
3. Learn probability relationships from historical transaction data.
4. Use the learned model to calculate the probability that a new transaction is fraudulent.
5. Test the model on unseen data to check **generalization**.
6. Update the model with new fraud patterns when more data becomes available.

Example:
Unusual location + unusual device + very high transaction amount → high fraud probability.

Advantage: Graphical models capture relationships between variables and can improve fraud prediction while handling uncertainty.